

IAM and Authentication

Series: ONBRD | **Notebook:** 2 of 10 | **Created:** December 2025

Setting Up Secure Access

Before inviting your team, configure authentication and permissions properly. This notebook covers SAML/SSO setup, API tokens, and the modern permission model.

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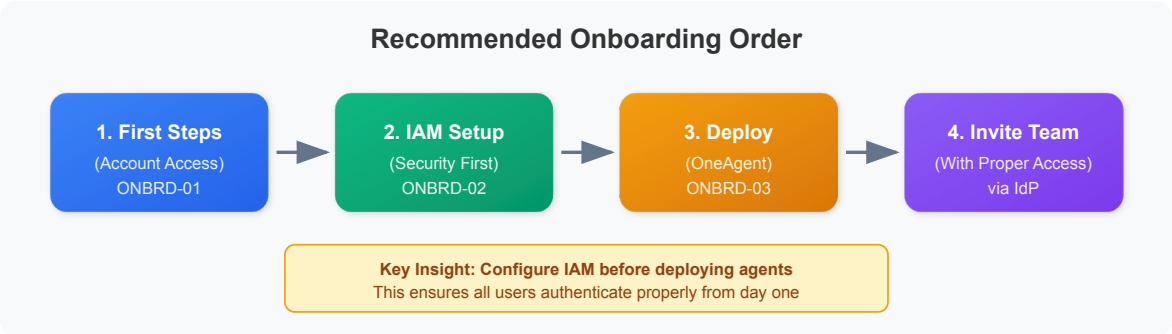
Prerequisites

- Account owner or admin access
- Identity provider details (for SSO)
- Understanding of your organization's access requirements

1. Why IAM First?

Setting up IAM before deploying OneAgent or inviting users ensures:

Benefit	Why It Matters
Consistent access	Users authenticate the same way from day one
Proper permissions	No accidental admin access for viewers
Audit trail	All access tied to corporate identity
Offboarding	Disabling IdP account revokes Dynatrace access
Compliance	Meet security requirements from the start

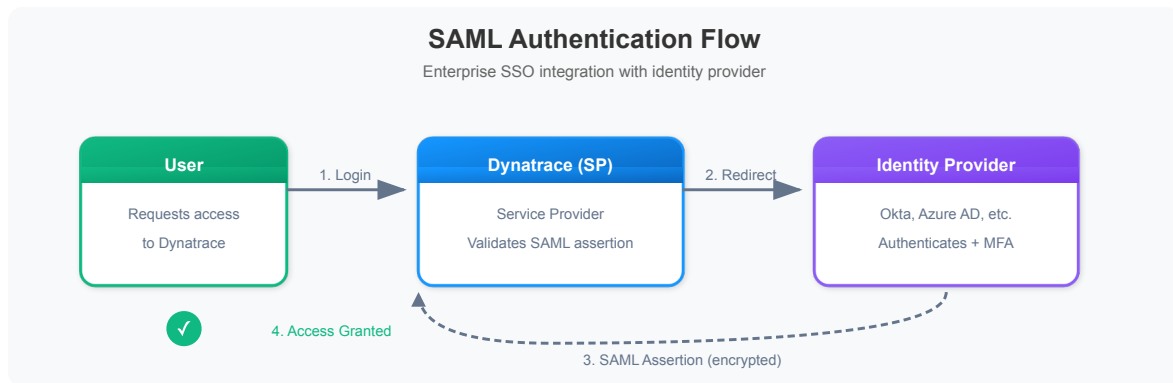


2. Authentication Options

Dynatrace supports multiple authentication methods:

Method	Description	Best For
Local Users	Built-in user accounts	Small teams, testing, break-glass accounts
SAML 2.0	Enterprise SSO integration	Most organizations
OIDC	OpenID Connect integration	Modern identity providers

Authentication Flow



Common Identity Providers

- **Azure Active Directory (Entra ID)** - Microsoft environments
- **Okta** - Cloud-native identity
- **OneLogin** - Enterprise identity
- **PingFederate** - On-premises/hybrid
- **Google Workspace** - Google-centric organizations

3. Configuring SAML SSO

Step 1: Access Account Management

Location: Account Management → Identity & access management → Single sign-on

You'll need these values for your IdP configuration:

Field	Description
Entity ID	Unique identifier for Dynatrace
ACS URL	Assertion Consumer Service URL
Relay State	Where users land after login

Step 2: Configure Your Identity Provider

In your IdP, create a new SAML application with:

1. **ACS URL** from Dynatrace
2. **Entity ID** from Dynatrace
3. **Name ID Format:** Email address (recommended)
4. **Required Attributes:**
 - `email` - User's email address
 - `firstName` - User's first name
 - `lastName` - User's last name

Step 3: Configure Dynatrace

Location: Account Management → Identity & access management → Single sign-on

1. Upload or paste the IdP metadata XML
2. Or manually configure:
 - **IdP Entity ID** - From your IdP
 - **SSO URL** - Login endpoint
 - **Certificate** - IdP signing certificate
3. Configure attribute mappings
4. Save and test

Step 4: Test SSO

Before enabling SSO for all users:

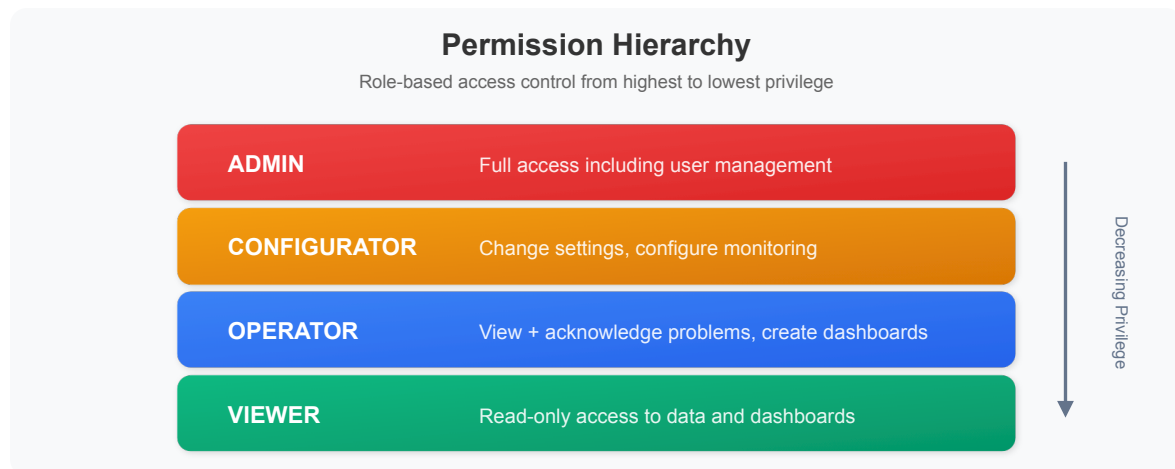
1. Open an incognito/private browser window
2. Navigate to your tenant URL (`https://{tenant-id}.apps.dynatrace.com`)
3. Select "Sign in with SSO"
4. Complete IdP authentication
5. Verify you land in Dynatrace with correct permissions

Warning: Keep at least one local admin account as a break-glass option in case SSO fails.

4. User Groups and Permissions

Permission Model

Dynatrace uses a policy-based access control model with the following permission levels:



Recommended Groups

Group	Role	Use Case
Platform Admins	Admin	Platform team, IAM management
SRE Team	Configurator	Workflow setup, monitoring config
Developers	Operator	Problem response, dashboards
Stakeholders	Viewer	Reports, read-only access

Creating Groups

Location: Account Management → Identity & access management → Groups

1. Click "Create group"
2. Name the group (e.g., "SRE-Team")
3. Add a description
4. Assign policies (permissions)
5. Optionally link to IdP groups for automatic membership

Scoping Access with Policies

The modern platform uses **policies** to control what users can access:

Policy Type	Purpose
Environment policies	Access to specific environments
Account policies	Account-level management
Data policies	Access to specific data (via segments)

Note: For data filtering, use **Segments** (covered in ONBRD-05) rather than the legacy Management Zones.

5. API Token and OAuth Management

The modern platform supports two authentication methods for API access:

API Tokens (for OneAgent, legacy integrations)

Location: Account Management → Access tokens

Purpose	Required Scopes
OneAgent Deployment	InstallerDownload
Metric Ingestion	metrics.ingest
Log Ingestion	logs.ingest
Read Entities	entities.read

OAuth 2.0 Clients (recommended for new integrations)

Location: Account Management → OAuth clients

OAuth 2.0 is the modern authentication method:

1. Create an OAuth client

2. Use client credentials flow to obtain bearer tokens
3. Tokens are short-lived and automatically rotated

Token Best Practices

Practice	Why
Minimal scope	Limit blast radius if compromised
Descriptive names	Know what each token is for
Expiration dates	Force rotation, reduce risk
Separate tokens per use	Revoke one without affecting others
Never commit to code	Use environment variables or secrets managers
Prefer OAuth for new work	Modern, short-lived tokens

Creating an API Token

1. Go to Account Management → Access tokens
2. Click "Generate new token"
3. Enter a descriptive name (e.g., "prod-oneagent-deployment")
4. Select required scopes
5. Set expiration (consider 90-365 days)
6. Click "Generate token"
7. **Copy the token immediately** - it won't be shown again

6. Verification Queries

After configuring IAM, verify your setup with these queries.

```
// Check recent audit log entries for user access
fetch logs, from: now() - 24h
| filter matchesPhrase(log.source, "audit")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Check for authentication events (if audit logs enabled)
fetch logs, from: now() - 7d
| filter matchesPhrase(content, "login") or matchesPhrase(content,
"authentication")
| fields timestamp, content
| sort timestamp desc
| limit 25
```

Manual Verification Checklist

Item	How to Verify
SSO working	Login via IdP in incognito window
Groups created	Check Account Management → Groups
Permissions assigned	Test with a viewer account
Break-glass account	Local admin login still works
API tokens	OneAgent deployment token ready

7. Next Steps

With IAM configured, you're ready to:

1. **ONBRD-03: Deploying ActiveGate** - Set up network routing (if needed for restricted networks)
2. **ONBRD-04: Deploying OneAgent** - Start collecting infrastructure data
3. Invite team members via your IdP
4. Create additional API tokens or OAuth clients as needed

IAM Tasks Before Moving On

- ☐ SAML/SSO configured and tested
- ☐ Break-glass local admin account documented
- ☐ User groups created for major roles
- ☐ OneAgent deployment token generated
- ☐ Token naming conventions established

Summary

In this notebook, you learned:

- Why IAM should be configured before deploying agents
 - Authentication options (Local, SAML, OIDC)
 - How to configure SAML SSO
 - Permission levels and group structure
 - API token and OAuth best practices
 - How to verify IAM configuration
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References

- [Identity and Access Management](#)
 - [SAML Configuration](#)
 - [OIDC Configuration](#)
 - [Access Tokens](#)
 - [OAuth Clients](#)
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